

BLAST Analysis of a Protein Sequence Using NCBI BLAST

Student Details :

Submitted by: Srishty Satybola

Course: B.Tech Biotechnology (2nd Year)

College: NIET, Greater Noida

Internship: CodeAlpha Bioinformatics Internship

Task: Task 1 – BLAST Analysis

Date: (30-6-26)

Objective:

The objective of this experiment is to retrieve a protein sequence from the NCBI or UniProt database and perform a BLAST analysis to identify homologous sequences. The study aims to compare sequence similarity, determine sequence identity, and analyze alignment scores to understand evolutionary relationships between proteins.

Introduction:

BLAST (Basic Local Alignment Search Tool) is a widely used bioinformatics tool that compares a query DNA or protein sequence against sequences stored in biological databases. It helps identify homologous sequences, predict protein function, and study evolutionary relationships. In this experiment, a protein sequence was downloaded from the UniProt (or NCBI) database and analyzed using NCBI BLAST.

Materials Used

- NCBI BLAST
- UniProt (or NCBI)
- Internet Browser
- Microsoft Word

Procedure

Number the steps, for example:

1. Selected a protein sequence from NCBI.

 **National Library of Medicine**
National Center for Biotechnology Information

Protein

FASTA

RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain; Contains: RecName: Full=LVV-hemorphin-7; Contains: RecName: Full=Spinorphin

UniProtKB/Swiss-Prot: P68871.2

[GenPept](#) [Identical Proteins](#) [Graphics](#)

>sp|P68871.2|HBB_HUMAN RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin; AltName: Full=Hemoglobin beta chain; Contains: RecName: Full=LVV-hemorphin-7; Contains: RecName: Full=Spinorphin

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTATLSELHCDKLHVDPENFRLLGNVLCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH



2. Downloaded the sequence in FASTA format.
3. Opened the NCBI BLAST website.
4. Pasted the sequence into the BLAST search box.

BLAST® » blastp suite Home

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

PDAVMGNPKVKAHGKKVLG
 AFSDGLAHLNLTGTFATLSLHCDKLHVDPENFRLLGNVLCVLAHFGKE
 FTTPVQAAAYQKVAVAN
 ALAHKYH

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database [?](#)

Organism [Add organism](#)

Optional [?](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Program Selection

Algorithm ☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

5. Executed the BLAST search.
6. Analyzed the top homologous sequences.
7. Recorded sequence identity, query coverage, E-value, and bit score.

Clusters producing significant alignments

Download

Select columns

Show

100

☒ select all 100 clusters selected

[GenPept](#)

[Graphics](#)

[Distance tree of results](#)

[Multiple alignment](#)

[MSA Viewer](#)

	Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
	Click the to see the cluster contents									
☒	1 member(s), 1 organism(s)	human	Hb Monza protein [Homo sapiens]	288	288	100%	1e-97	86.47%	170	XEF57200.1
☒	1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AAA35597.1
☒	7 member(s), 6 organism(s)	dog_coyote_wolf_fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
☒	183 member(s), 75 organism(s)	placentals	hemoglobin subunit delta [Ailuropoda melanoleuca]	273	273	100%	6e-92	89.80%	147	NP_001291814.1
☒	18 member(s), 17 organism(s)	placentals	hemoglobin subunit beta-like [Ailuropoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001291812.1
☒	7 member(s), 5 organism(s)	bats	hemoglobin subunit beta [Rhynchonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM8815943.1
☒	1 member(s), 1 organism(s)	giant_panda	hypothetical protein PANDA_015769 [Ailuropoda melanoleuca]	275	446	100%	3e-91	90.48%	258	EFB18430.1
☒	1 member(s), 1 organism(s)	large_flying_fox	TPA_globin A2 [Pteropus vampyrus]	272	272	100%	7e-91	88.44%	193	SAI82279.1
☒	4 member(s), 4 organism(s)	primates	PREDICTED: LOW QUALITY PROTEIN: hemoglobin subunit...	270	270	100%	7e-91	89.80%	147	XP_011830554.1
☒	37 member(s), 25 organism(s)	cellular_organisms	hemoglobin beta-like protein, partial [Apibacter sp. B3889]	269	269	100%	3e-90	87.76%	162	WIP_221411036.1
☒	20 member(s), 12 organism(s)	carnivores	hemoglobin subunit beta isoform X3 [Neomonachus schauinsl]	267	267	100%	1e-89	86.39%	147	XP_044775197.1

Parameter	Result
Query Sequence	Hemoglobin Beta Protein
Database Used	NCBI Protein BLAST (BLASTP)
Top Homologous Sequence	Hb Monza protein (<i>Homo sapiens</i>)
Query Coverage	100%
Percentage Identity	86.47%
E-value	1e-97
Max Score	288
Total Score	288
Accession Number	XEF57200.1
Protein Length	170 amino acids

Conclusion:

The BLAST analysis successfully identified homologous protein sequences with significant similarity. The obtained sequence identity, bit score, and E-value confirmed that the selected protein shares evolutionary relationships with proteins present in the database. This experiment demonstrated the usefulness of BLAST in sequence comparison and functional analysis.